

New Mangalore Port Authority

Visit Briefing — Questions, Data Asks & Project Status

PROJECT

Maritime Port Intelligence Platform (MPIP) — NMPA use case

PURPOSE OF THIS VISIT

Validate our cost, tariff and vessel-data assumptions directly with NMPA, and identify which of our project's remaining gaps NMPA can realistically help close.

HOW TO USE THIS DOCUMENT

One page per question. Each page gives the background, what we already know/have built, and a boxed line to literally ask on the day. The last page is a one-page cheat sheet for the room.

This briefing was prepared from an internal audit of our own codebase and datasets — every claim below about what is “real” versus “synthetic” in our platform, and every number quoted from the NMPA Scale of Rates (SOR effective 01.05.2026–30.04.2027), was checked line-by-line against source before being written down here.

Their business runs on tariffs — how does that affect our project?

What NMPA's business actually is

NMPA is a Major Port Authority operating entirely on a government-regulated Scale of Rates (SOR) — currently the 2026-27 SOR, effective 01 May 2026 to 30 Apr 2027. It covers vessel-related charges (port dues, berth hire, pilotage, anchorage, tug hire, shifting) and cargo-related charges (wharfage across 60+ commodity lines, container box rates, transit storage). Rates are ceiling levels set under the Tariff Authority for Major Ports (TAMP) framework and re-indexed every 1 May to 60–100% of the Wholesale Price Index (WPI).

Why this matters directly to us

Our Port Cost Estimator is the one module that actually touches NMPA's real commercial business. Our forecasting and analytics screens are advisory; a cost estimate is something an agent or shipper could act on. Accuracy here isn't optional in the way it might be elsewhere in the platform.

What we've already built

- We transcribed the published SOR directly into our cost engine — wharfage, berth hire, port dues, pilotage, anchorage, tug hire, shifting, and transit storage.
- Example checks that match the SOR line-for-line: crude oil via SPM wharfage ■34.89/MT (coastal) / ■34.90/MT (foreign); dry-bulk berth hire ■0.07602/GRT/hr (coastal), subject to a ■262.83/hr minimum.

Where we're still exposed

- **Static table, manual refresh.** The SOR re-indexes every 1 May; our numbers go stale unless someone manually re-transcribes the new PDF each year.
- **Some components have no fixed numeric rate** in the SOR itself (e.g. cruise-vessel composite charges, container scanning, LDB mandatory user charges) — it just says “as per separate order,” so we can't compute those without that order.
- **Zero real invoices to check against.** Every number is transcribed from the tariff, not verified against an actual bill.

ASK THEM AT THE VISIT

- *Do you keep the SOR (or a subset) in a structured/spreadsheet form, or only as the published PDF?*
- *Could someone in billing or operations sanity-check 3–4 of our sample cost estimates against a real invoice?*
- *Who owns the annual WPI re-indexation each 1 May — is there a notice we can subscribe to?*

QUESTION 2 OF 4

Which APIs do we actually need — and which can NMPA realistically give us?

Not every gap in our platform is NMPA's to fill. The useful exercise is separating data NMPA generates and owns from data NMPA merely consumes from someone else — we should only be asking for the former.

Likely theirs to give — worth asking

- **Vessel-call register** (arrival / berth allotted / departure per ship, IMO number, cargo handled) — HIGH VALUE. Almost certainly exists internally for Port Control / berth planning, even if never published. This single feed would let us build real vessel-stay and berth-occupancy predictions instead of the assumption-based proxy we run today (see Q5b).
- **Live / periodic berth occupancy status** — medium value, likely already exists in their berth planning system.
- **Structured tariff feed** — low technical value but removes our manual PDF-to-code transcription step (see Q1).
- **Circulars & Trade Notice feed** — we already scrape their public “Circular and Trade Notice” page; worth asking if an RSS/API/bulk export exists instead of HTML scraping.

Probably NOT theirs to give — wrong department

- **Cargo manifest / import-export documentation** — that's Customs (ICEGATE) data, not the Port Authority's.
- **Live global AIS / vessel positions at sea** — that's a commercial vendor's product (MarineTraffic, Spire). NMPA's own in-harbour tracking is a related but separate ask — see Q3.
- **Commodity market prices, trade-lane statistics** — government trade-statistics or market-data agencies (DGCIS, World Bank), not the port itself.

ASK THEM AT THE VISIT

- *Does Port Control keep a structured vessel-call log we could get periodic exports from?*
- *Is there an RSS/API for the Circular & Trade Notice page, or should we keep scraping it?*
- *Who's the right contact for each of the above — Traffic, IT, or Harbour Master's office?*

We're missing AIS and currently download vessel data by hand — what should we actually ask for?

Where we are today

Our “live vessel” screen is a fixed, scripted 25-ship list — not live, not historical, a placeholder. Getting anything real today means looking up vessels one at a time by hand; there is no bulk historical export in our current workflow.

Why we shouldn't just buy commercial AIS

MarineTraffic/Spire-grade satellite AIS is a paid subscription priced for fleet operators, and most of what it would show us — ships already inside NMPA's harbour limits — is redundant with data the port already collects for its own safety and berth-allocation operations, via its Vessel Traffic Management System (VTMS) / Port Control. We'd be paying to re-derive something NMPA already has.

The actual ask, reframed

We don't need to ask NMPA to “give us AIS.” We need to ask whether Port Control / VTMS can export the vessel-call log it already keeps: vessel name / IMO number, ETA/ATA, berth assigned, ATB (berthing time), ATD (departure time), cargo type and quantity handled. Even a periodic bulk CSV/Excel extract — not live streaming — would be transformative: it is exactly the missing input for real stay-duration and berth-occupancy modeling.

Secondary ask: historical depth

If 2–3 years of vessel-call history could be shared (sensitive commercial fields redacted if needed), we could backtest a real turnaround-time model the same way we already backtest our cargo forecasts, and validate it against NMPA's own committed performance standard — 37-hour average vessel turnaround, 7-hour average pre-berthing time (SOR Annexure-II).

ASK THEM AT THE VISIT

- *Does Port Control / VTMS maintain a per-vessel call log (arrival, berth, departure)?*
- *Could a historical bulk extract — even monthly, even redacted — be shared for a student/research project?*
- *Who's the right department to ask — Traffic, IT, or Harbour Master's office?*

Beyond an API, what else could NMPA help with?

Validation, not just data

Asking someone in operations or billing to eyeball a handful of our cost estimates and flag anything that looks off is cheaper than any dataset, and directly de-risks the one module that touches their real business (Q1).

Domain review of our assumptions

Our what-if scenario simulator (vessel-arrival shift %, demand shift %, weather-delay days) currently runs on assumed elasticities, not fitted ones — we say so explicitly in our own documentation. Twenty minutes with a traffic/planning officer sanity-checking whether those assumed numbers are in the right ballpark would materially improve credibility even without any new data.

PCS1x / Port Community System

India's major ports increasingly route cargo and vessel data through the unified Port Community System (PCS1x) rather than bespoke exports. Worth asking directly whether NMPA participates and whether a research/student integration is possible — this could be a single doorway to several of the Q2/Q3 asks instead of separate one-off requests.

A named point of contact

For the annual tariff re-indexation (Q1), and for who to notify if our platform is ever demoed internally and someone spots an error — an ongoing relationship is worth more than a one-time data dump.

Digital / “Smart Port” initiatives

Ask if there's a digital transformation or Smart Port cell (many major ports have one under the Sagarmala programme) — that's the natural home for a student project like ours and the fastest route to a yes on any of the above.

ASK THEM AT THE VISIT

- *Could ops/billing spend a few minutes sanity-checking our cost estimates against a real invoice?*
- *Does NMPA participate in PCS1x, and is a student/research integration possible?*
- *Is there a digital transformation / Smart Port cell we should be talking to?*

What in our platform is genuinely real today?

Before asking NMPA for anything, we audited our own platform end-to-end. This page and the next are an honest summary — useful both for us and as a credibility signal if NMPA staff ask “what exactly are you showing me.”

Cargo forecasting — REAL, validated

RandomForest model trained on 1,354 real monthly NMPA commodity-traffic records (2021–2026), backtested out-of-sample with measured accuracy (WAPE $\approx$ 17%, $\approx$ 83% accuracy vs. a seasonal-naive baseline).

Vessel-call forecasting & anomaly detection — REAL, partially validated

Same real dataset, RandomForest / IsolationForest models. Anomaly detection is real ML but unsupervised — there's no labeled ground truth to check it against.

Port cost estimator — REAL tariff data, not yet invoice-validated

Built directly from the published SOR (see Q1) — real numbers, not invented — but never checked against an actual NMPA bill.

Berth / facility routing — REAL NMPA facility data

Uses real berths.csv / berth_capacity.csv NMPA facility data with rule-based commodity-to-berth matching. Our own audit caught and fixed a substring-matching bug that had been silently routing crude oil, LPG, coal and containers to Berth 1 (a shallow general-cargo berth) because “Berth 1” matched as a substring of “Berth 10” through “Berth 17.”

NMPA Circulars & Trade Notices — not built, but confirmed easy

Confirmed scrapable from the public “Circular and Trade Notice” page with no NMPA dependency — a fully achievable addition using only public data.

What we can't solve yet, and the specific reason why

Vessel stay duration / berth occupancy prediction — BLOCKED

Our only NMPA data is monthly aggregated tonnage, not per-vessel call records — there is no dwell-time signal to learn from at all. Solvable only with the vessel-call log described in Q3.

Real-time vessel tracking (AIS) — BLOCKED

Blocked by cost (commercial AIS is a paid subscription) and by redundancy (NMPA's own VTMS already has this for ships in port). Solvable via the Q3 ask, or, failing that, honestly relabeled as a simulated feed rather than the current fake static fleet list.

Trade intelligence (trade lanes, commodity prices, “opportunities”) — BLOCKED, but not on NMPA

We have no origin/destination or market-price dataset at all — these numbers are currently randomly generated on every request. Solvable independently of NMPA, via DGCIS / Ministry of Commerce trade statistics and World Bank commodity price data — an ask for us to pursue on our own, not something to raise at the port visit.

Digital twin, executive KPI dashboard, AI copilot, security/SOC module — NOT YET REAL

Currently fully synthetic — randomly generated on every request, including fabricated revenue figures and a fake “847,203 postgresql_records” value. Not “blocked” so much as never started for real. Needs either a real data source behind each screen, or the UI needs to visibly say “simulated” as clearly as our own code comments already do.

What-if scenario simulator — mechanism real, calibration missing

The simulator itself runs correctly, but its elasticity constants are documented assumptions, not fitted to real data — same root cause as the vessel-stay gap above: no real congestion/delay dataset exists yet to calibrate against.

One-page cheat sheet — what to literally ask

MUST-ASK

- 1. “Does Port Control / VTMS keep a per-vessel call log (arrival, berth, departure)? Could a historical bulk extract be shared for a student/research project?” (Q3 — *highest impact, unlocks vessel-stay and berth-occupancy prediction.*)
- 2. “Is there a digital transformation / Smart Port / PCS1x integration point we should be talking to instead of, or in addition to, Traffic/IT?” (Q4 — *could be a single doorway to several asks.*)
- 3. “Could someone in billing or operations sanity-check a handful of our cost estimates against a real invoice?” (Q1/Q4 — *cheapest, highest-credibility ask.*)

NICE-TO-ASK

- “Is the SOR maintained anywhere in structured/spreadsheet form, or only as the published PDF?” (Q1)
- “Is there an RSS/API for the Circular & Trade Notice page, or should we keep scraping it?” (Q2)
- “Who owns the annual WPI re-indexation announcement each 1 May, and is there a way to subscribe to it?” (Q1)
- “Could a traffic/planning officer spend 20 minutes reviewing our what-if simulator's assumptions for realism?” (Q4)

WHAT NOT TO ASK NMPA FOR — redirect elsewhere

- Global AIS / MarineTraffic-grade live positions — a commercial vendor's product, not the port's.
- Trade-lane and commodity-price data — DGCIS / World Bank, not the port.
- Import-export cargo manifests — Customs (ICEGATE), not the Port Authority.

Prepared from an internal, line-by-line audit of the project codebase and the NMPA Scale of Rates (effective 01.05.2026–30.04.2027). Every “real” / “synthetic” claim on the Project Status pages traces to a specific function or dataset in the codebase.